Network Modeling – AA 2016/2017
written test – July 13, 2017 – part A (90 minutes)

E1 Consider a network node that works as follows. If there is no traffic, the node alternates between a sleep state for an exponential duration of average T and an awake state for an exponential duration of average βT. When in the awake state, the node can receive, whereas in the sleep state it cannot. With probability α, during its stay in the awake state the node receives a message request. In this case, *at the end* of the awake period a link is established and the node starts receiving the message, whose average duration is γT. With probability $1-\alpha$, instead, nothing is received during the awake time, and the node goes back to sleep. Develop and solve a semi-Markov model for the node, and in particular:

(a) Consider the three states sleep (S), listening (L) and receiving (R), and determine the matrix of the transition probabilities of the embedded Markov chain, and draw its transition diagram.

(b) Determine the average times associated to the visits to each state, μ_S, μ_L, μ_R.

(c) Find an expression for the fraction of time the node spends in each of the three states, and find its numerical value for $\alpha = 0.5, \beta = 0.3, \gamma = 0.2$.

E2 Consider a Go-Back-N protocol over a two-state Markov channel, where the steady-state probability that the channel is in the bad state is 0.1 and the average number of consecutive good slots is 100. The packet error probability is 1 for a bad slot and 0 for a good slot, respectively. The round-trip time is $m = 2$ slots, i.e., a packet that is erroneous in slot t will be retransmitted in slot $t+2$.

(a) Compute the throughput that could be obtained if packets were directly transmitted over the channel without using any protocol

(b) Compute the throughput of the Go-Back-N protocol for an error-free feedback channel

(c) Compute the throughput of the Go-Back-N protocol for a feedback channel subject to iid errors with probability 0.1.

E3 Consider a Markov chain X_n with the following transition matrix (states are numbered from 0 to 2):

$$P = \begin{pmatrix} 0.2 & 0.6 & 0.2 \\ 0.6 & 0.2 & 0.2 \\ 0 & 0 & 1 \end{pmatrix}$$

(a) Draw the transition diagram, and find the probability distribution of X_1, X_2 and X_{1000}, given $X_0 = 0$

(b) Let $W_{ij}^{(n)} = E\left[\sum_{k=0}^{n-1} I\{X_k = j\} \middle| X_0 = i\right]$ be the average number of visits to state j during the first n time slots, given that the chain starts in state i. Compute $\lim_{n\to\infty} W_{0j}^{(n)}$ for $j = 0, 1, 2$.

(c) Compute the average duration of the transient evolution of the chain, i.e., the time index at which the chain is absorbed.

E4 Consider two independent Poisson processes $X_1(t)$ e $X_2(t)$, where $X_i(t)$ is the number of arrivals for process i during $[0, t]$. The average number of arrivals per unit time of the two processes is $\lambda_1 = 0.5$ and $\lambda_2 = 1$, respectively.

(a) Compute $P[X_1(1) = 1 | X_1(2) + X_2(2) = 3]$ and $P[X_1(2) + X_2(2) = 3 | X_1(1) = 1]$

(b) Compute $P[X_1(3) = 2 | X_1(2) + X_2(2) = 0]$ and $P[X_1(3) = 2 | X_1(2) + X_2(2) = 1]$

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T1 For a renewal process, state precisely (also providing a formal proof) what is the value of

$$\lim_{t\to\infty} \frac{N(t)}{t}$$

T2 Prove that if states i and j of a Markov chain communicate and i is recurrent, then j is also recurrent.

T3 For a Poisson process of rate λ, prove that the interarrival times are iid exponential with mean $1/\lambda$